

CM0845 Logic

Propositional Logic: Satisfiability

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2016-1

Propositional Logic: Satisfiability

Remark

The reference for this section is Ben-Ari (2012, § 2.5).

Satisfiability, Validity, Unsatisfiability and Falsifiability

Definitions

Let $\varphi \in \text{PROP}$.

- (i) φ is **satisfiable** iff $\llbracket \varphi \rrbracket_v = 1$ for some interpretation v .
In this case, v is called a model for φ .
- (ii) φ is **valid** (a tautology), denoted $\models \varphi$, iff $\llbracket \varphi \rrbracket_v = 1$ for all interpretations v .
- (iii) φ is **unsatisfiable** iff it is not satisfiable, that is, if $\llbracket \varphi \rrbracket_v = 0$ for all interpretations v .
- (iv) φ is **falsifiable**, denoted $\not\models$, iff it is not valid, that is, if $\llbracket \varphi \rrbracket_v = 0$ for some interpretation v .

Satisfiability, Validity, Unsatisfiability and Falsifiability

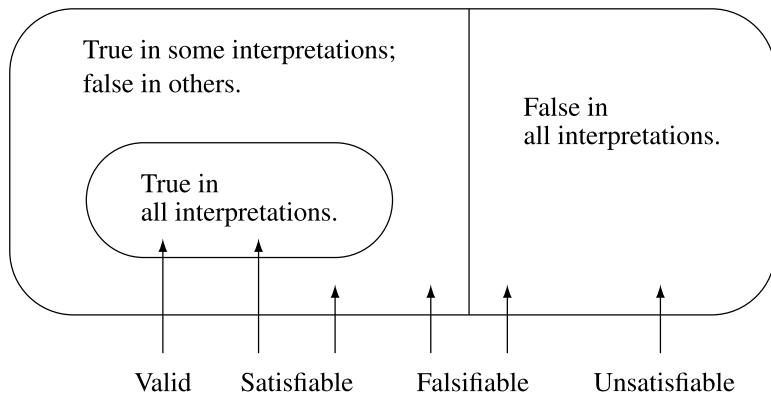


Figure 2.6 of (Ben-Ari 2012).

Satisfiability, Validity, Unsatisfiability and Falsifiability

Theorem (Ben-Ari (2012), Theorem 2.39)

Let $\varphi \in \text{PROP}$.

- (i) The proposition φ is valid if and only if $\neg\varphi$ is unsatisfiable.
- (ii) The proposition φ is satisfiable if and only if $\neg\varphi$ is falsifiable.

Satisfiability of a Set of Propositions

Definitions

Let $\Gamma = \{\varphi_1, \dots\}$ be a set of propositions.

- (i) Γ is **satisfiable** iff there exists an interpretation v such that $\llbracket \varphi \rrbracket_v = 1$ for all $\varphi_i \in \Gamma$.
In this case, v is a model of Γ .
- (ii) Γ is **unsatisfiable** iff for every interpretation v , there exists an $\varphi_i \in \Gamma$ such that $\llbracket \varphi \rrbracket_v = 0$.

Satisfiability of a Set of Propositions

Definitions

Let $\Gamma = \{\varphi_1, \dots\}$ be a set of propositions.

- (i) Γ is **satisfiable** iff there exists an interpretation v such that $\llbracket \varphi \rrbracket_v = 1$ for all $\varphi_i \in \Gamma$.
In this case, v is a model of Γ .
- (ii) Γ is **unsatisfiable** iff for every interpretation v , there exists an $\varphi_i \in \Gamma$ such that $\llbracket \varphi \rrbracket_v = 0$.

Example

Prove that if Γ is unsatisfiable and for some i , the proposition φ_i is valid, then $\Gamma - \{\varphi_i\}$ is unsatisfiable (Ben-Ari 2012, Exercise 2.15, p. 46).

References



Mordechai Ben-Ari [1993] (2012). Mathematical Logic for Computer Science. 3rd ed. Springer (cit. on pp. 2, 4–7).